

MIBI single-cell data
(54 patients, 54,989 cells,
29 protein markers + xy coordinates)



Lineage gating
(per-marker 80th-percentile
positivity + hierarchical rules)



Per-patient feature engineering
(composition, density, PD-L1,
NND proximity, Ripley's K)



Locked analytical dataset
(54 patients $\times$ 44 features)



Leave-one-out cross-validated
random-forest classification + univariate group comparisons